Revolutionising B.Tech

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Biodiversity

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- Introduction to biodiversity
- Definition: genetic, species and ecosystem diversity
- Biogeographical classification of India
- Value of biodiversity;
- Biodiversity at global, national and local levels
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- Hot-spots of biodiversity
- Threats to biodiversity conservation of biodiversity
- In-situ and ex-situ conservation of biodiversity.

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Aim

To equip students in the fundamentals and understanding of biodiversity and their use in preserving the diversity of species.

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Objective

a. To preserve the diversity of species.

b. Sustainable utilization of species and ecosystem.

c. To maintain life-supporting systems and essential ecological processes.

d. To preserve the continuity of food chains.

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Biodiversity ???

- Biodiversity refers to every living thing, including plants, bacteria, animals, humans and their interactions

- Biodiversity comes from two words Bio meaning life and diversity meaning variability.

- Scientists have estimated that there are around 8.7 million species of plants and animals in existence. However, only around 1.2 million species have been identified and described so far,

Why is Biodiversity important?

Biodiversity is essential for the processes that support all life on Earth, including humans. Without a wide range of animals, plants and microorganisms, we cannot have the healthy ecosystems that we rely on to provide us with the air we breathe and the food we eat.

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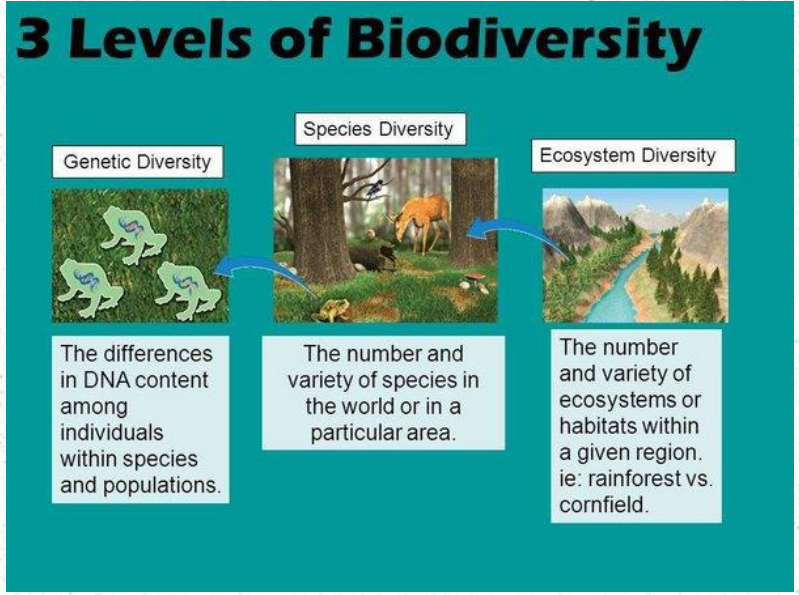

Levels of biodiversity

Biodiversity is usually explored at three levels - genetic diversity, species diversity and ecosystem diversity. These three levels work together to create the complexity of life on Earth.

Genetic diversity

Genetic diversity is the variety of genes within a species. Each species is made up of individuals that have their own particular genetic composition. This means a species may have different populations, each having different genetic compositions.

Example:

Different breeds of dogs as a result of selective breeding is one of the best example of genetic diversity. There are more than 50,000 varieties of rice and more than a thousand varieties of mangoes found in India.

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Species diversity

- Species diversity is the variety of species within a habitat or a region. Some habitats, such as rainforests and coral reefs, have many species. Others, such as salt flats or a polluted stream, have fewer.
- Diversity is greatest when all the species present are equally abundant in the area.
- Example: No. of plants, animals, birds in an area.

Ecosystem diversity

Ecosystem diversity is the variety of ecosystems in a given place. An ecosystem is a community of organisms and their physical environment interacting together. An ecosystem can cover a large area, such as a whole forest, or a small area, such as a pond.

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BIOGEOGRAPHICAL CLASSIFICATION OF INDIA

→ The Bio-geographic Zone
→ The Biotic Province
→ The Land Region
→ The Biome
→ Climate and Topography
→ Variety of Flora and Fauna
→ Rich heritage of Biological Diversity

Biogeography is the study of the distribution of species (biology), organisms, and ecosystems in geographic space

India has a rich heritage of natural diversity. India ranks fourth in Asia and tenth in the world amongst the top 17 mega-diverse countries in the world. India harbours nearly 11% of the world's floral diversity comprising over 17500 documented flowering plants, 6200 endemic species, 7500 medicinal plants and 246 globally threatened species in only 2.4% of world's land area.

India stand at 10th position among plant rich nations of the world.

- A bio-geographic Zone is a large distinctive unit of similar ecology, biome representation, community and species.
- The biotic province is secondary unit within bio-geographic zone, giving weight to particular community separated by dispersal barriers or gradual change in environmental factors for example North West and Western Himalayas either side of the Sutlej River.
- A land region is a tertiary set of units within a province, indicating different land forms, e.g. Aravalli Mountain and Malwa Plateau in Gujrat- Rajwara Province.
- A biome is an ecological unit, not a bio-geographical unit. It can be defined as various similar ecosystems throughout the world grouped together.

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India's Major Biographic Habitats

S. No.	Biogeographic Zone	Biotic Province	Total Area (Sq. Km)
1	Trans Himalayan	Upper Regions	186200
2	Himalayan	North-West Himalayan West Himalayas Central Himalayas East Himalayas	6900 720000 123000 83000
3	Desert	Kutch Thar Ladakh	45000 180000 NA
4	Semi Arid	Central India Gujarat – Rajwara	107600 400400
5	Western Ghats	Malabar Coasts Western Ghat Mountains	59700 99300
6	Deccan Peninsula	Deccan plateau south Central plateau Eastern plateau Chhota nagpur Central highlands	378000 341000 198000 217000 287000
7	Gangetic Plain	Upper gangetic plains Lower gangetic plain	206400 153000
8	North-East India	Brahmaputra valley North-eastern hills	65200 106200
9	Islands	Andaman islands Nicobar islands Lakshadweep islands	6397 1930 180
10	Coasts	West coast East coast	6500 6500

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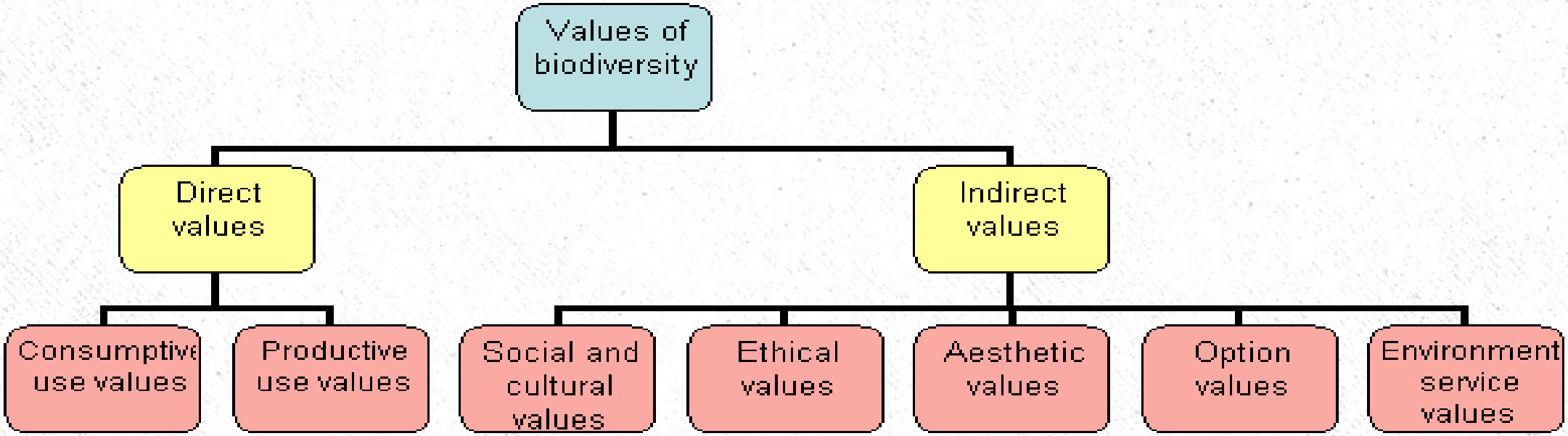
Values of Biodiversity…

- It is in terms of commercial utility, ecological services, social & aesthetic Values
- Given by McNeely et al. 1990.



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Consumptive use Value

- In the form of food, drugs, fibre pulp, fuel, wood fall under this group
- Food: 80,000 edible plants species are there.
  - About 90% of present day food crops have been domesticated from wild plants
  - A large number of wild animals are also source of food.
- Drugs & Medicines:75% world's population is dependent on plants or plant extracts for medicines
  - Isabgol- cure for bowel disorder obtained from husk of *Plantago.*
  - *Penicillin*- A wonder drug – used as Antibiotic – Penicillium
  - *Tetracyclin*- from bacterium ,*Quinine* – from Cinchona Tree- medicine for Malaria, *Digitalin*- from foxglove- for Heart Ailments ,*vinblastin* & *vincristine* – Anti-Cancer Alkaloids
- Fuel: big source of fuel wood. Firewood collected by individuals are direct consumptive value

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Productive Use Values

- These are commercially usable values where the product is marketed & sold
- Musk Deer- production of Musk , Silk Worm- Silk, wool from sheep, lac from insects.
- Many industries are dependent on these products, e.g. pulp & paper industry, silk industry, ivory work, pearl industry, leather industry.
- Problems--- trading of products from endangered species.
- E.g smuggling of fur, horns, tusk, snake skin, cat skins.
- Rich biodiversity are exploited.

Social Values

Values associated with social life, customs, religion and pyscho-spiritual aspects of the people

Many plants are considered holy & sacred. Eg. Tulsi, mango, peepal etc.

Social life and customs are woven with surrounding wildlife

Cow, snake, owl, bull have special value in biodiversity

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Ethical Value

- The ethical and moral value of biodiversity is the principle that humans should abide by to ensure the conservation of all forms of life around us.
- the ethical use of the biodiversity is that all organisms must be allowed to live in their own Habitat.
- The ethical reasoning behind biodiversity conservation aims mainly to conserve other organisms that co-inhabit earth because each organism is important in its own right even if we cannot derive direct economic benefits from them.

Aesthetic Value

- It is related to the beauty of Biodiversity
- The pleasure, excitement and visual peace of any area
- Concept of Eco-tourism and willingness to pay are gaining grounds, leading to monetary estimate for aesthetic value of biodiversity

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Option Value

- This includes the potentials of biodiversity that are presently unknown and need to be explored.
- This biological resources will be of important in future if not today
- Eg. Marine animals – anti cancer drugs

Ecosystem Value

- It refers to ecosystem services, it is a non-consumptive value.
- Functions of ecosystems- nutrient cycling, nitrogen fixation, cycling of water. Etc.

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Global Biodiversity….

- Currently about 1.9 million species are known,
- Thought to be a significant underestimate of the total number of species (as of 2010).
- 0-30 million insects;
- 5-10 million bacteria;
- 1.5 million fungi;
- 1 million mites
- 3,21,212 – Plants
  - 1,367,555 animals, including: 1,305,250 invertebrates ,62305 vertebrates

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Did u know this?

- 50 to 80% of Biodiversity is in rain forests.
- Deserts show minimum biodiversity
- Many species are identified everyday…
- List goes on increasing
- Marine biodiversity is extremely high.
- 1,25,000 flowering plant species are in Tropical forests
- Roughly 1.5 million species are known till date which may be just 15% of actual number.

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Biodiversity at National Level (Indian)

- India is one of the 12 mega diversity countries of the world.
- 2.4% of the land area, accounting for 7-8% of the species of the world.
- 10th among plant rich countries
- 11th in number of endemic species of higher vertebrates
- 6th in terms of centres of diversity

Ministry of Environment and Forests records (as of 2000)

- 47000 plants: 7% of global.
- 91000 species of mammals: 6.5% of global.
- Large numbers have ENDEMIC - ORIGINATED in India
- 5000 flowering plants..
- 340 corals

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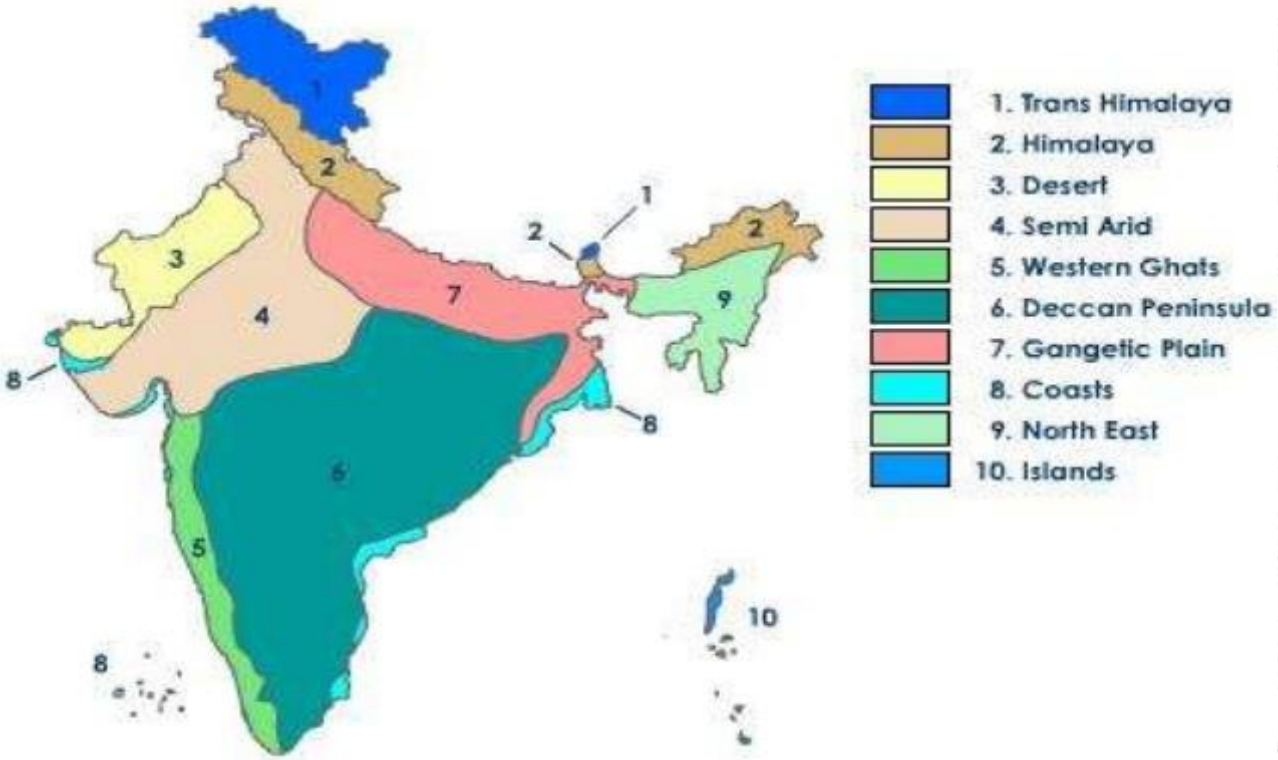
Biogeographical Regions of India

10 Biogeographic Zones of India



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Biodiversity at Regional or Local

- This type of biodiversity can be characterized in richness of four types based on their spatial distribution
  - Point Richness- species that are found at one single point
  - Alpha (α)- richness) – number of species found in small homogeneous area. Here Physical factor is strongly corealted.
  - Eg. 100 species of tunicates in Arctic region, 400 species in temperate waters & 600 species in tropical seas
  - Beta (β)– richness) - refers to rate of change in species composition across different habitats. Cumulative no. of species increases in heterogeneous habitats.
  - The No. of Ant species in local area will be 10 but it will be more than 200 no of species in equatorial area, due to increase in habitats.
  - Gamma (γ)- Richness – refers to the rate of change across large composition gradients.

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Hotspots of Biodiversity

- A **biodiversity hotspot** is a biogeographic region with a significant reservoir of biodiversity that is under threat from humans.
- Areas which exhibit high species richness as well as endemism are termed as Hotspots of Biodiversity.
- Myers introduced this term, at that time 25 Hotspots were identified out of which 2 were in India. Later 9 were added more bringing a total to 34.
- About 40% terrestrial & 25 % vertebrate are endemic found in this hotspots
- After tropical rain forests the second highest number endemic species are found in Mediterranean
- These hotspots are threatened by human activities. More than 1 billion people most whom are desperately poor people, live in these areas.
- Measures protecting these areas should be planned.

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INDIA AS MEGA DIVERSITY NATION

India is one of the 12 mega diversity countries in the world.

- → Australia
- → Brazil
- → China
- → Columbia
- → Ecuador
- → The United States
- → India
- → Indonesia
- → Madagascar
- → Mexico
- → Peru
- → The democratic Republic of Congo

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India – as a Mega-Diversity Nation

- Endemism- species which are restricted only to a particular area. Eg. 50% Lizards are endemic to India. Western Ghats are the site of maximum endemism.
- Center of origin: Nearly 5000 species of flowering plants, 166 species of cropping plants and 320 species of wild relatives of cultivated crops have their origin in India.
- Marine Diversity: India has 7500 Km long coastline having (mangroves, estuaries, coral reefs, black waters) rich biodiversity.
  - More than 340 species of corals are found in India
  - Rich in Mollusks, crustaceans, polychaetes, corals
  - 93 major wetlands, Large Forest Cover.

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→The ministry of environment and forests, Govt. of India records 47000 species of plants and 81000 species of animals i.e., 7% and 6.5% of the global flora and fauna.

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Distribution of species in some major groups of flora and fauna in India

Group wise Species Distribution			
Plants	Number	Animals	Number
Bacteria	850	Lower Groups	9979
Fungi	23000	Mollusca	5042
Algae	2500	Arthopoda	57525
Bryophytes	2564	Pisces	2546
		Amphibia	428
Pteridophytes	1022	Reptiles	1228
Gymnosperms	64	Birds	204
Angiosperms	15000		
		Mammals	372

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Endemism – Species which are restricted to

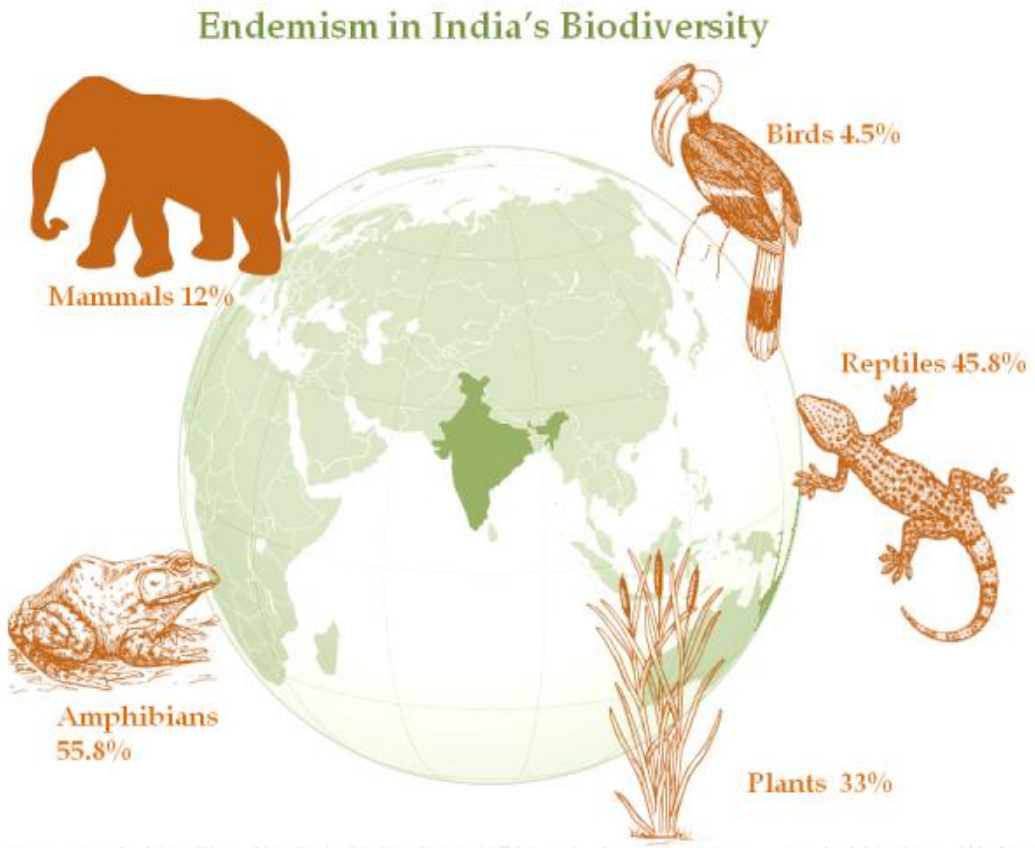
a particular area are known as endemic.

About 62% of amphibians and 50% lizards

are endemic to India.

Western Ghats are the site of maximun

endemism.



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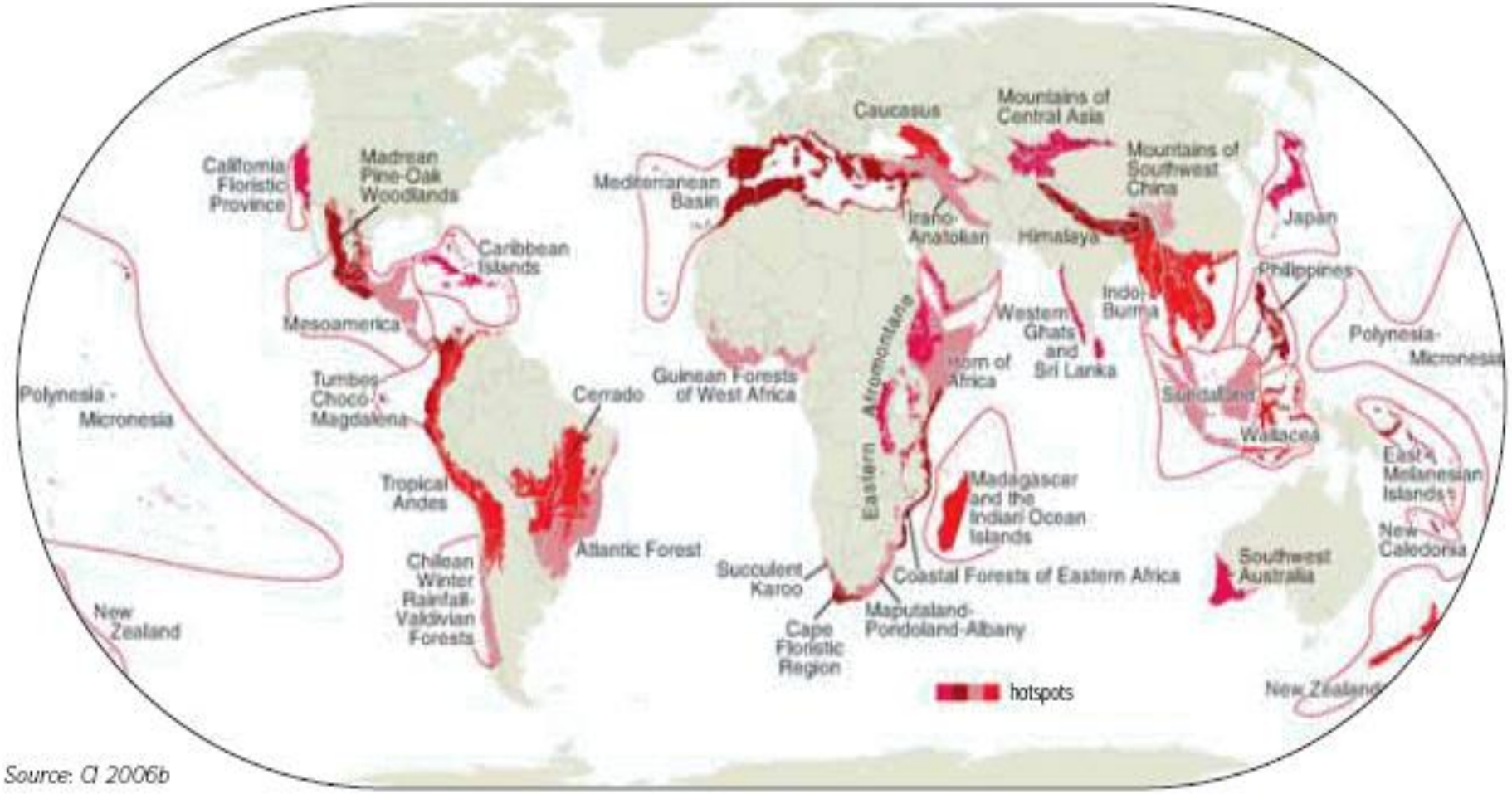
Biodiversity Hotspots

Figure 4: Biodiversity hotspots



Source: CI 2006b

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List of Hotspots

Sr. No	Name of Hotspots	Sr. No	Name of Hotspots
1	Tropical Andes	19	Indo-Burma
2	Mesoamerican Forests	20	South Central China
3	Caribbean	21	Western Ghats
4	Brazil's Atlantic Forest	22	South-Western Australia
5	Choc/Darien of Panama Western Ecuador	23	New Caledonia
6	Brazil's Cerrado	24	New Zealand
7	Central Chile	25	Polynesia/Micronesia
8	California Floristic Province	26	The Madrean Pine-Oak woodlands
9	Madagascar	27	Maputaland-Pondoland Albany
10	Eastern Arc & Coastal forest of Tanzania/ Kenya	28	The eastern Afromontane
11	Western African Forest	29	The Horn Africa
12	Cape Floristic Province	30	The Irano-Anatolian
13	Succulent Karoo	31	The Muntains of Central Asia
14	Mediterranean Basin	32	Eastern Himalaya
15	Caucasus	33	Japan
16	Sundaland	34	East Melanesiaan Islands
17	Wallacea		
18	Phillipines		

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Threats to Biodiversity

→Extinction or elimination of a species is a natural process of evolution.

→However, the rate of loss of species has been a slow process.

→The process of extinction has become particularly fast in the recent years of human civilization.

→One of the estimates by E O. Wilson puts the figure of extinction at 10,000 species per year.

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Threats to biodiversity

- **Anthropogenic causes**
  - Loss of habitat
  - Poaching
  - Man- Animal conflict
  - Introduction of exotic varieties
  - Pollution
  - Global warming and Climate change

- **Natural causes**
  - Narrow geographical area
  - Low population
  - Low breeding rate
  - Natural disasters

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Loss of habitat

- Continuous increase in human population and escalating demand for our natural resources
- Conversion of forested land to agriculture.
- Loss of habitat to grazing land, industries, roads and cities.
- Habitat loss leads to the formation of isolated, small, scattered populations.
- These small populations are increasingly vulnerable to inbreeding, which causes loss of gene pool, high infant mortality and susceptible to environmental changes, which all may lead to extinction of the species.

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Poaching

- Hunting for various body parts.
- Illegal trade of skins, tusks, hair, horns
- Many animals fall prey to various traps
- International market very active
- Tibet, China
- Tiger skins, elephant tusks etc
- Cost is very less.

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Biodiversity Conservation

- Considering the rich diversity and the threats conservation of this vast biodiversity is very important
- 2 major approaches
- **In-situ** conservation (at the place)
- **Ex-situ** conservation (out of the place)

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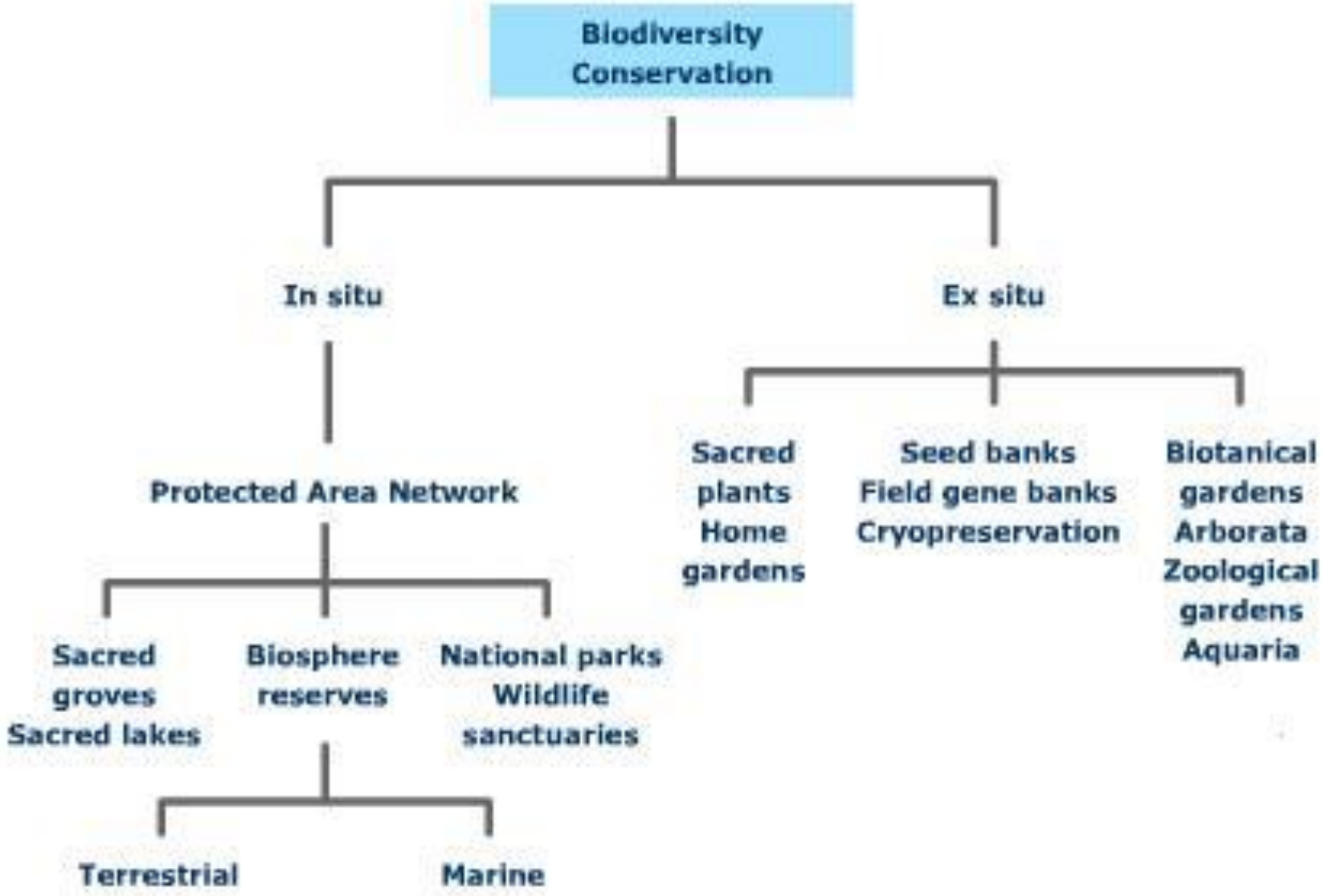
Conservation Measures



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In Situ Conservation

- At present there are 7 major Biosphere Reserves, 80 National Parks, 420 Wild-Life Sanctuaries and 120 botanical gardens in India covering 4% of geographic area.
- The ***Biosphere Reserves*** conserve some representative ecosystems as a whole for long term *in situ* conservation. In India we have Nanda Devi, Nokrek Manas, Sunderbans, Gulf of Mannar, Nilgiri, Great NIcobars and Similipal Biosphere reserves.

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- A **National Park** is an area dedicated for the conservation of wild-life along with its environment. It is also meant for enjoyment through tourism but without impairing the environment.
- Each National Park usually aims at conservation specifically of some particular species of wild-life along with others.

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- Wild-Life Sanctuaries are protected areas where killing, hunting, shooting or capturing of wild-life is prohibited except under the control of highest authority.
- Private Ownership rights are permissible and forestry operations are also permitted to an extent that do not affect the wild-life adversely.

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The different advantages of in situ conservation are as follows:

(a) It is a cheap and convenient way of conserving biological diversity.

(b) It offers a way to preserve a large number of organisms simultaneously, known or unknown to us.

(c) The existence in natural ecosystem provides opportunity to the living organisms to adjust to differed' environmental conditions and to evolve in to a better life form.

The only disadvantage of in situ conservation is that it requires large space of earth which is often difficult because of growing demand for space. The protection and management of biodiversity through in situ conservation involve certain specific areas known as protected areas which include national parks, Sanctuaries and Biosphere reserves.

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Ex Situ Conservation

- This type of conservation is mainly done for conservation of crop varieties, the wild relatives of crops and all the local varieties with the main objective of conserving the total genetic variability of the crop species for future crop improvement programs.
- In India there are 4 important gene bank/seed bank facilities:

→ NBPGR

→ NBAGR

→ NFPTCR

→ NBFGR

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Ex situ conservation Institutes

- **Some important areas under these conservation are:**
- (i) Seed gene bank,
- (ii) Field gene bank;
- (iii) Botanical gardens;
- (iv) Zoos.

- 196 zoos, 34 botanical gardens
- Gene bank/ seed bank Facilities:
- National Bureau of Plant Genetic Research (NBPGR)
- National Bureau of Animal Genetic Research (NBAGR)
- National Facility for Plant Tissue Culture Repository(NFPTCR)

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Some important areas under these conservation are:

(i) Seed gene bank,

(ii) Field gene bank;

(iii) Botanical gardens,

(iv) Zoos.

The strategies for ex-situ conservations are:

(i) Identification of species to be conserved.

(ii) Adoption of Different ex-situ methods of conservation.

(i) Long-term captive breeding and propagation for the species which have lost their habitats permanently.

(ii) Short-term propagation and release of the animals in their natural habitat

(iii) Animal translocation

(iv) Animal reintroduction

(v) Advanced technology in the service of endangered species.

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The different advantages of ex-situ conservation are:

(a) It gives longer life time and breeding activity to animals.

(b) Genetic techniques can be utilised in the process.

(c) Captivity breed species can again be reintroduced in the wild.

Some disadvantages of this method are:

(a) The favourable conditions may not be maintained always.

(b) Mew life forms cannot evolve.

(c) This technique involves only few species.

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THANK YOU

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Aim

To equip students in the fundamentals and understanding of biodiversity and their use in preserving the diversity of species.

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Assessments

Question 1:

Which of the following statements best describes biodiversity?

A) **Biodiversity refers to the variety of species and ecosystems on Earth.**

B) Biodiversity is the total number of individuals within a population.

C) Biodiversity is the measure of the physical size and shape of an organism.

D) Biodiversity is the ability of an organism to adapt to its environment.

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Assessments

Question 2:

Which of the following strategies is a key approach for the conservation of biodiversity?

A) Habitat destruction and fragmentation.

B) Overexploitation of natural resources.

C) Introduction of non-native species.

D) Establishment and management of protected areas.

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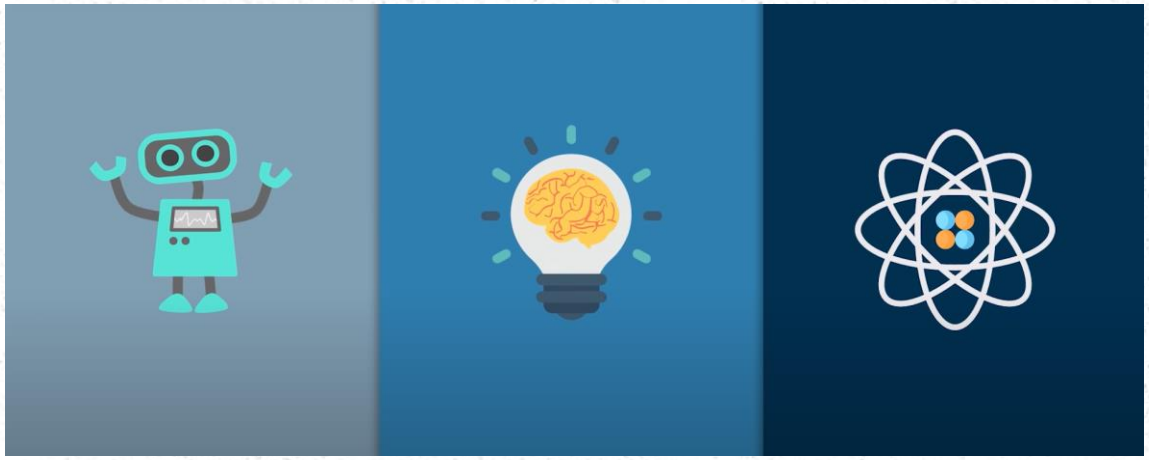
Did You Know?



50 to 80% of Biodiversity is in rain forests.

Deserts show minimum biodiversity

Many species are identified everyday…

List goes on increasing

Marine biodiversity is extremely high.

1,25,000 flowering plant species are in Tropical forests

Roughly 1.5 million species are known till date which may be just 15% of actual number.

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Summary

Outcomes:

a. Students will realize that people are dependent on intact habitats that sustain the various organisms we need to produce food, medicines, clothing, and other materials.
b. Students will learn about certain species' roles in an ecosystem.
c. Students will discover that life can be found almost everywhere on earth.

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